

Results: Citation Network

## RQ4: Citation Geometry I

Resolving each record's references against the corpus yields an intra-corpus citation graph (built and analyzed with NetworkX (Hagberg et al. 2008)) of 2234 nodes and 8,623 edges across 1409 connected components, with a graph density of 0.17% and a mean in-degree of 3.9. Of 38,489 total outgoing references, 22.4% resolve to another record inside the corpus — a resolution rate that reflects how self-contained the retrieved slice of the literature is rather than the underlying citation density of any single work.

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The citation network has 1416 communities (detected by modularity optimization), a maximum in-degree of 163 (the most-cited paper within the corpus), and a maximum out-degree of 143 (the paper that cites the most other corpus members).

Centrality scores (PageRank (Page et al. 1999) and HITS) and modularity-based community detection (Clauset et al. 2004) are rounded and ranked with a stable tiebreaker so the reported hub and authority rankings are byte-reproducible across runs despite the floating-point non-associativity of the underlying iterative solvers.

**Table 4. Top 5 papers by PageRank.**

# Centrality Analysis I

| Rank | DOI                                | PageRank |
|------|------------------------------------|----------|
| 1    | 10.1177/026988110001400107         | 0.036943 |
| 2    | 10.1212/wnl.54.5.1166              | 0.026949 |
| 3    | 10.1523/jneurosci.21-05-01787.2001 | 0.013668 |
| 4    | 10.1523/jneurosci.20-22-08620.2000 | 0.011349 |
| 5    | 10.4088/jcp.v61n0510               | 0.008136 |

**Table 5. Top 5 authority papers (HITS).**

# Centrality Analysis I

| Rank DOI |                                    | Authority |
|----------|------------------------------------|-----------|
| 1        | 10.1038/sj.npp.1301534             | 0.017700  |
| 2        | 10.1007/s00213-002-1250-8          | 0.016311  |
| 3        | 10.1124/jpet.106.106583            | 0.015144  |
| 4        | 10.1523/jneurosci.21-05-01787.2001 | 0.014734  |
| 5        | 10.1001/jama.2009.351              | 0.014481  |

**Table 6. Top 5 hub papers (HITS).**



# Centrality Analysis I

| Rank DOI |                               | Hub      |
|----------|-------------------------------|----------|
| 1        | 10.3389/fnins.2021.656475     | 0.011968 |
| 2        | 10.1016/bs.apha.2023.10.006   | 0.011596 |
| 3        | 10.1038/sj.npp.1301534        | 0.010886 |
| 4        | 10.1080/08897077.2019.1700584 | 0.010562 |
| 5        | 10.1007/s00213-013-3232-4     | 0.009896 |

The most influential paper by PageRank (DOI [10.1177/026988110001400107](https://doi.org/10.1177/026988110001400107)) is a foundational work that anchors the citation structure — its high authority score confirms it is frequently cited by other corpus members. Hub papers, which cite many other corpus members, serve as integrative reviews or meta-analyses that connect disparate threads of the literature.

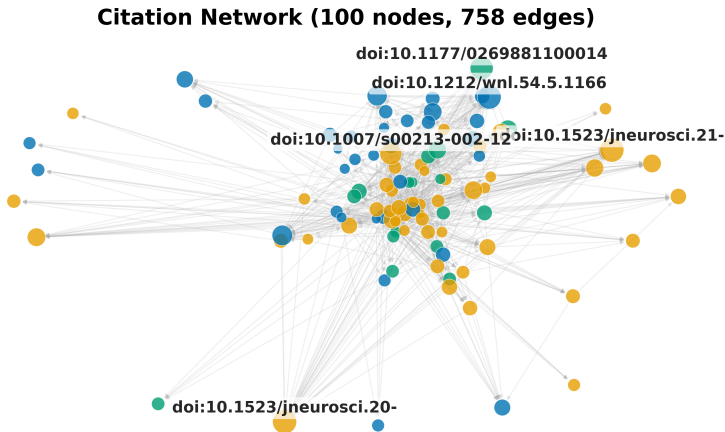


Figure 1: Citation network for Modafinil. Nodes represent papers; directed edges represent citation links. Node colours indicate community membership (1416 communities detected by modularity optimization). Layout uses a spring-based algorithm with a fixed seed for reproducibility.

## Centrality Analysis II

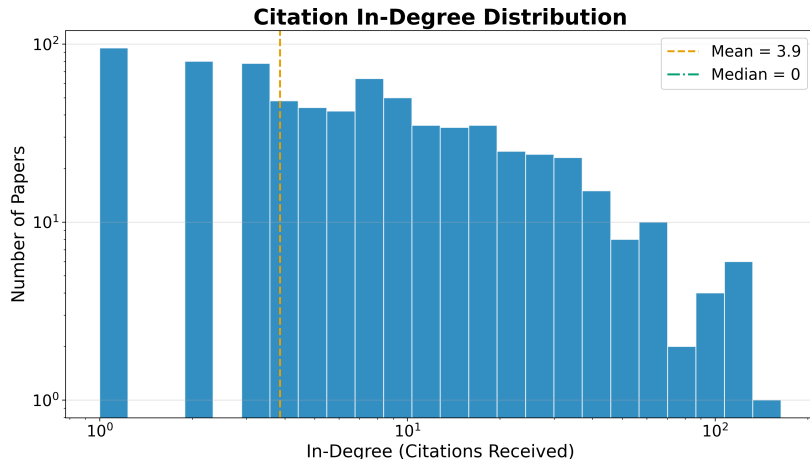


Figure 2: Degree distribution for the Modafinil citation network. The histogram shows the frequency of each in-degree value on a log-linear scale, revealing the heavy-tailed structure characteristic of citation networks.

The heavy-tailed degree distribution is characteristic of citation networks: a small number of highly-cited papers anchor the structure, while the long tail of low-degree nodes represents newer or peripheral works. The low graph density (0.17%) reflects the sparsity of intra-corpus citation links — most papers cite works outside the retrieved slice, which is expected for a max-results-capped retrieval.

Beyond PageRank and HITS, the network analysis computes betweenness centrality (which papers bridge different communities), closeness centrality (which papers are near all others), degree assortativity (do highly-cited papers cite other highly-cited papers?), and average clustering coefficient (how tightly knit are local neighborhoods).

The degree assortativity coefficient is  $-0.0598$ , and the average clustering coefficient is  $0.1029$ . A negative assortativity indicates that highly-cited papers tend to cite less-cited papers (disassortative mixing), which is typical of citation networks where review papers (high in-degree) cite many primary studies (low in-degree).

**Table 7. Top 5 papers by betweenness centrality.**

## Advanced Network Metrics I

| Rank DOI |                                  | Betweenness |
|----------|----------------------------------|-------------|
| 1        | 10.1038/sj.npp.1301534           | 0.005234    |
| 2        | 10.4088/jcp.v67n0406             | 0.003451    |
| 3        | 10.1016/j.neuropharm.2012.07.011 | 0.002530    |
| 4        | 10.1002/14651858.cd006788.pub3   | 0.002289    |
| 5        | 10.2165/00003495-200868130-00003 | 0.001894    |



Papers with high betweenness centrality serve as bridges between different topical communities in the citation network — their removal would fragment the graph into disconnected components. These bridging papers are often review articles or methodological papers that connect disparate research threads.